

Conversational MCP Tools & Real-World Taxpayer Painpoints Stress & Accuracy Report

Date: 2026-08-21
Evaluation Target: Single Dedicated GPU (NVIDIA RTX A6000 - GPU 2, `CUDA_VISIBLE_DEVICES="2"`)
Evaluated Stack: Model Context Protocol (MCP 2026-07-28 Stateless JSON-RPC Spec) + FastAPI Frontend
Conversational Endpoints (`/v1/chat` , `/v1/chat/stream` , `/v1/escalate` , `/v1/tax/*`)
Core Dimensions: Response Structure Conformance, MCP Tool Precision, Multi-Turn Real-World Painpoint Resolution, and Concurrency Stress (\$c = 10 lto 250\$).

1. Executive Summary

This report provides comprehensive empirical verification of the **conversational architecture, response structure adherence, and Model Context Protocol (MCP) tool activations** exposed via the frontend APIs under realistic multi-turn taxpayer scenarios.

Key Highlights

- MCP Direct Tool Activation & Precision:** **100.0% arithmetic precision** and schema conformance across all registered statutory tax calculators (`check_vat_registration` , `calculate_paye` , `calculate_customs_duty` , `calculate_rental_tax` , `calculate_withholding` , `escalate_to_human`).
- Real-World User Painpoints Resolution:** **100.0% completion rate** across 6 high-impact Uganda taxpayer painpoints (EFRIS VAT threshold, PAYE progressive brackets, Customs import valuation, Rental tax, Withholding tax, and Distressed taxpayer escalation).
- Response Structure Conformance:** **100.0% strict JSON schema conformance** (`reply` , `citations` , `confidence` , `session_id` , `disclaimer`) under all concurrency levels.
- Conversational Concurrency Scaling (\$c = 10 lto 100\$):** Sustained **34.0 – 44.9 RPS** with **0% error rate** and sub-second median latencies.
- Extreme Concurrency Stress (\$c = 250\$, 500 requests):** **100.0% structure accuracy** at **48.2 RPS** with zero dropped connections.
- Traffic Spike Surge (\$c = 250\$ in 50ms):** Processed 250 burst requests at **49.4 RPS** with **0 circuit breaker trips**.
- High-Volume Conversational Soak (1,500 Continuous Queries):** Processed 1,500 continuous multi-turn queries in **30.75s (48.8 RPS)** with **0.0 MiB memory leak**.
- Hardware Isolation & Scoped Cleanup:** Dedicated execution on GPU 2; process cleanup strictly scoped to `~/.Mpairwe7` with zero interference with external cluster workloads.

2. MCP Tool Activation & Statutory Arithmetic Precision Matrix

Tool Identifier	Statutory Domain	Sample Test Payload	Output Key Verified	Arithmetic Precision	Latency (ms)	Sta
<code>check_vat_registration</code>	VAT Act (Cap 349) / FY2026-27	<code>annual_turnover:</code> UGX 350,000,000	<code>registration_required:</code> <code>true</code> (Threshold: 300M)	100.0%	0.155 ms	PA

Tool Identifier	Statutory Domain	Sample Test Payload	Output Key Verified	Arithmetic Precision	Latency (ms)	Stability
calculate_paye	Income Tax Act (Cap 340) Sch 4	monthly_gross: UGX 4,500,000 (Resident)	paye: UGX 1,238,250.00	100.0%	0.149 ms	PA
calculate_customs_duty	EAC Common External Tariff	cif_value: UGX 50,000,000, duty: 25%	duty: UGX 12.5M, vat: UGX 11.25M	100.0%	0.069 ms	PA
calculate_rental_tax	Income Tax Act s.5 (Individual)	annual_gross_rent: UGX 36,000,000	tax: UGX 3,981,600.00 (12% over 2.82M)	100.0%	0.040 ms	PA
calculate_withholding	Income Tax Act s. 119 (Services)	payment_type: "services", UGX 20,000,000	withholding_tax: UGX 1,200,000.00 (6%)	100.0%	0.031 ms	PA
escalate_to_human	Tax Procedures Code (SLA Queue)	priority: "urgent", reason: "Dispute"	ticket_id: UUID (Dispatched to queue)	100.0%	1.681 ms	PA

3. Real-World Taxpayer Painpoint Resolution via Frontend APIs (/v1/chat)

ID	Real-World User Painpoint	Tax Domain	Conversational Intent & Resolution	API Latency	Resolution Status
PP-01	EFRIS E-Invoicing & Mandatory VAT Registration	VAT & EFRIS	Retailer turnover UGX 350M exceeds statutory threshold of UGX 300M (FY2026-27); clarified immediate registration requirement.	2,239.6 ms	RESOLVED
PP-02	Employer Monthly PAYE Progressive Bracket Breakdown	PAYE	Calculated progressive tiered deductions across all statutory brackets for UGX 4.5M gross salary.	619.2 ms	RESOLVED
PP-03	Customs Commercial Cargo Import Duty & Port Valuation	Customs	Computed combined Customs Duty (25%) and Import VAT (18%) for commercial consignment at entry port.	614.4 ms	RESOLVED
PP-04	Residential & Commercial Rental Income Tax Assessment	Rental Tax	Assessed landlord tax at 12% on rent exceeding statutory threshold of UGX 2,820,000.	588.2 ms	RESOLVED
PP-05	Withholding Tax (WHT) Source Deduction on Consultancy	WHT	Computed 6% domestic service withholding tax deduction for accounting invoice clearance.	602.1 ms	RESOLVED
PP-06	Distressed Taxpayer Human Officer Escalation	Staff Queue	Dispatched urgent priority escalation ticket with SLA timer to human officer queue for frozen account dispute.	615.3 ms	RESOLVED

4. Conversational Concurrency Scaling Matrix (\$c = 10 \to 100\$)

Concurrency (\$c\$)	Requests Evaluated	Total Duration (s)	Container Throughput (RPS)	Structure Conformance (%)	Median Latency (p50)	Latency p95	Error Rate	Status
\$c = 10\$	40	1.176s	34.0 RPS	100.0%	247.7 ms	489.1 ms	0.0%	Optimal SLA
\$c = 25\$	50	1.205s	41.5 RPS	100.0%	468.2 ms	812.4 ms	0.0%	Linear Scaling
\$c = 50\$	100	2.591s	38.6 RPS	100.0%	1,029.8 ms	1,745.2 ms	0.0%	Medium Load
\$c = 100\$	200	4.454s	44.9 RPS	100.0%	1,940.4 ms	3,012.8 ms	0.0%	High Load

5. Extreme Stress, Traffic Surge Spike & High-Volume Soak

A. Extreme Concurrency Stress (\$c = 250\$, 500 Requests)

- **Throughput:** 48.2 RPS (500 requests completed in 10.37s).
- **Structure Conformance:** 100.0% (0 errors).
- **Latency:** p50 = 4,874.1 ms, p95 = 6,215.3 ms.

B. Instantaneous Traffic Spike Surge (\$c = 250\$ in 50ms)

- **Burst Profile:** Step surge from idle to 250 concurrent worker threads in 50ms.
- **Throughput:** 49.4 RPS (250 requests completed in 5.06s).
- **Reliability:** 100.0% structure conformance with 0 circuit breaker trips.

C. High-Volume Conversational Soak (1,500 Continuous Queries)

- **Total Workload:** 1,500 continuous multi-turn conversational tool queries under \$c = 50\$.
- **Throughput:** 48.8 RPS (completed in 30.75s).
- **Structure Conformance:** 100.0% across all 1,500 requests.
- **Memory Footprint:** 0.0 MiB heap or VRAM leak.

6. Single-GPU Hardware Telemetry & Scoped Cleanup

Metric	Measured Value	Operational Safety Verification
Target GPU	NVIDIA RTX A6000 (GPU 2)	CUDA_VISIBLE_DEVICES="2"
Total Hardware Capacity	49,140 MiB (48.0 GiB)	10,752 CUDA Cores
Active Peak VRAM Consumption	5,905 MiB (12.01%)	42,770 MiB (87.99%) free headroom
Post-Test Resource Cleanup	CLEAN (Scoped strictly to ~/Mpairwe7)	0 lingering processes; external jobs untouched

7. Artifacts & Source References

- Benchmark Runner: `scripts/test_mcp_conversational_painpoints_stress.py`
- Scoped Cleanup Tool: `scripts/cleanup_gpu_processes.py`
- Raw Metrics JSON: `Results/metrics/mcp_conversational_painpoints_report.json`
- Linked GitHub Issue: [Issue #304](#)